Heat-engine efficiency

antocciphysics | TEACHER KEY | CPA

Learning target: Calculate useful output and rejected energy for a heat engine.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 6; 3 course days later.*
Same average per particle; total translational energy is 3 times as large.

02. Explain a worked example (worked)
100=25+75 J; useful-work fraction 25%.

03. Look and explain (model)
Balances; 30% efficiency.

04. Build the solution (practice)
25%; 150 J.

05. Use the idea (reason)
Motor converts electrical work, not a cyclic heat-engine input; different constraints and input energy forms.

06. Choose a method (mixed) *Review: Unit 4, Day 19; 12 course days later.*
Both become 1/16 of their original values; both scale with v^2.

07. Exit check (exit)
40%; 60 J.

08. STRETCH: Apply it another way (stretch)
10 J unaccounted; could be stored energy or another transfer, or inconsistent measurements.

Heat-engine efficiency

antocciphysics | TEACHER KEY | HONORS

Learning target: Calculate useful output and rejected energy for a heat engine.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 6; 3 course days later.*
Rms speed becomes 3 times as large because mean kinetic energy is proportional to the square of rms speed.

02. Explain a worked example (worked)
100=25+75 J; useful-work fraction 25%.

03. Represent and interpret (model)
Balances; 30% efficiency.

04. Solve and justify (practice)
Max 50%, max 400 J; actual 30%.

05. Analyze the reasoning (reason)
Motor converts electrical work, not a cyclic heat-engine input; different constraints and input energy forms.

06. Choose a method (mixed) *Review: Unit 4, Day 19; 12 course days later.*
Coupled energy becomes 4/6≈0.6667 of the original; multiply the mass and fraction factors.

07. Exit check (exit)
450 J; 30%.

08. STRETCH: Apply it another way (stretch)
Bound 50%; actual 40%, below the bound. Rejected heat 600 J if steady cycle.